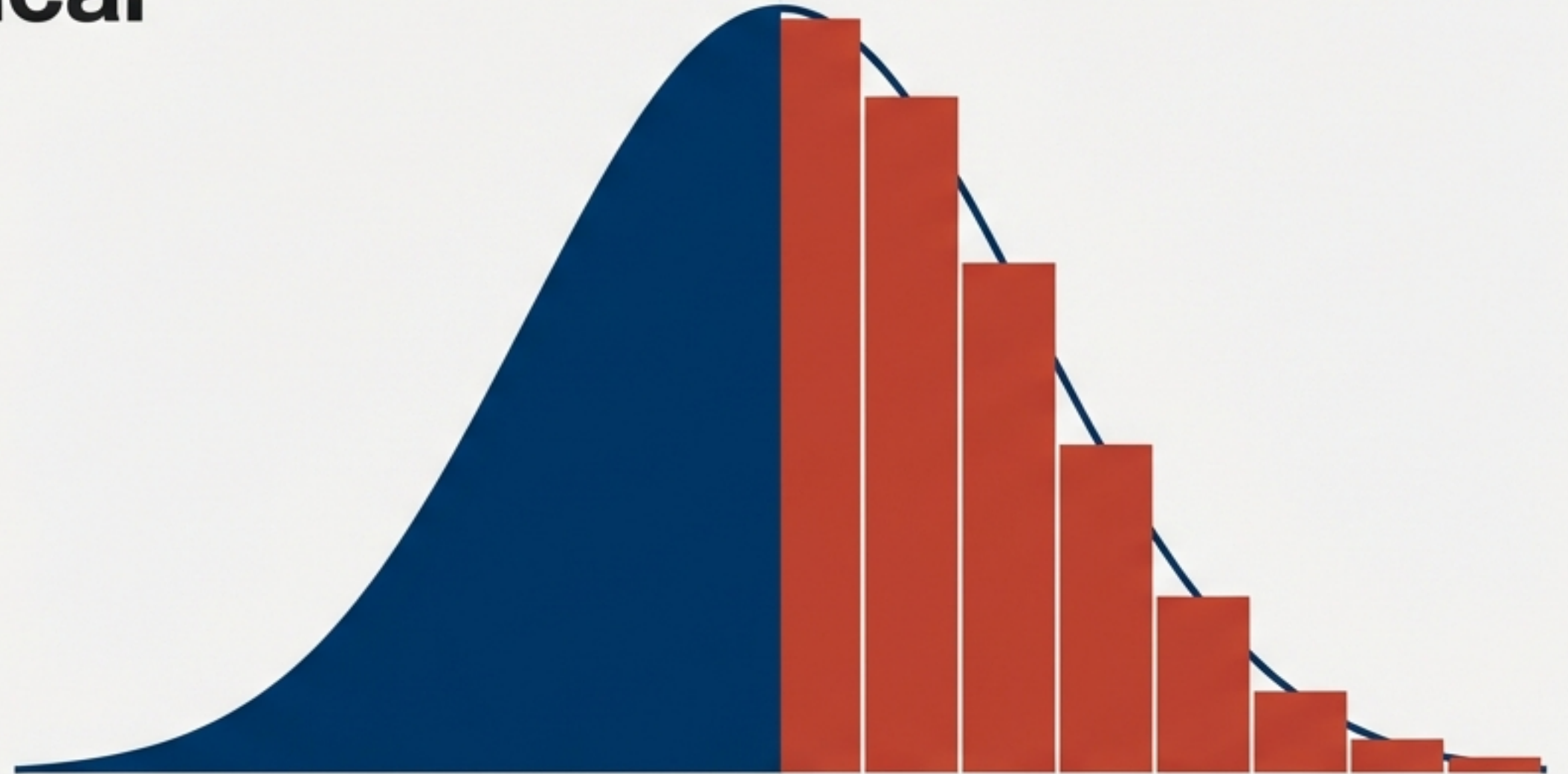


Modelling Uncertainty: A Guide to Statistical Distributions

A toolkit for understanding PDF, CDF, Z-Scores, and the Poisson Distribution.

Data comes in two fundamental forms: things we measure and things we count. To interpret the noise, we need specific mathematical lenses. This deck explores the core mechanics of Continuous and Discrete distributions.



The Two Landscapes of Data

Continuous Data



Measured values over a range
(e.g., height, temperature, time).

Primary Tools:

PDF & CDF

Discrete Data



Discrete Data. Countable events
in a fixed interval (e.g., typos per
page, emails per hour). Primary Tool:

Poisson Distribution

The nature of your data dictates the shape of your probability.

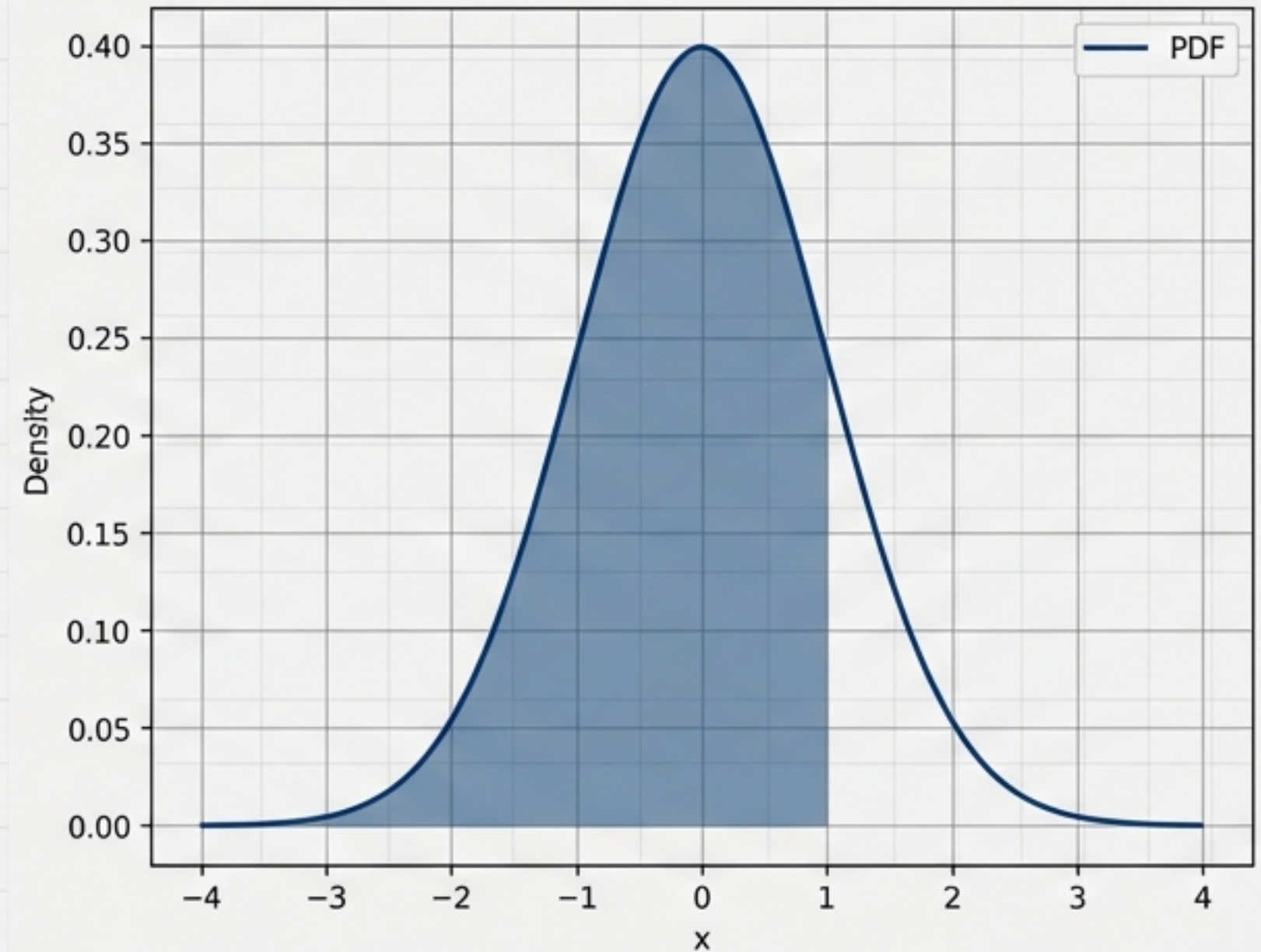
The Shape of Likelihood: Probability Density Function (PDF)

The PDF, denoted as $f(x)$, describes the likelihood of a random variable taking on a specific value.

Critical Nuance: For continuous variables, the probability of any exact value is 0. Instead, we calculate probability over an interval (the area under the curve).

Properties:

- Always ≥ 0
- Total area under the curve = 1



$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Accumulating Probability: Cumulative Distribution Function (CDF)

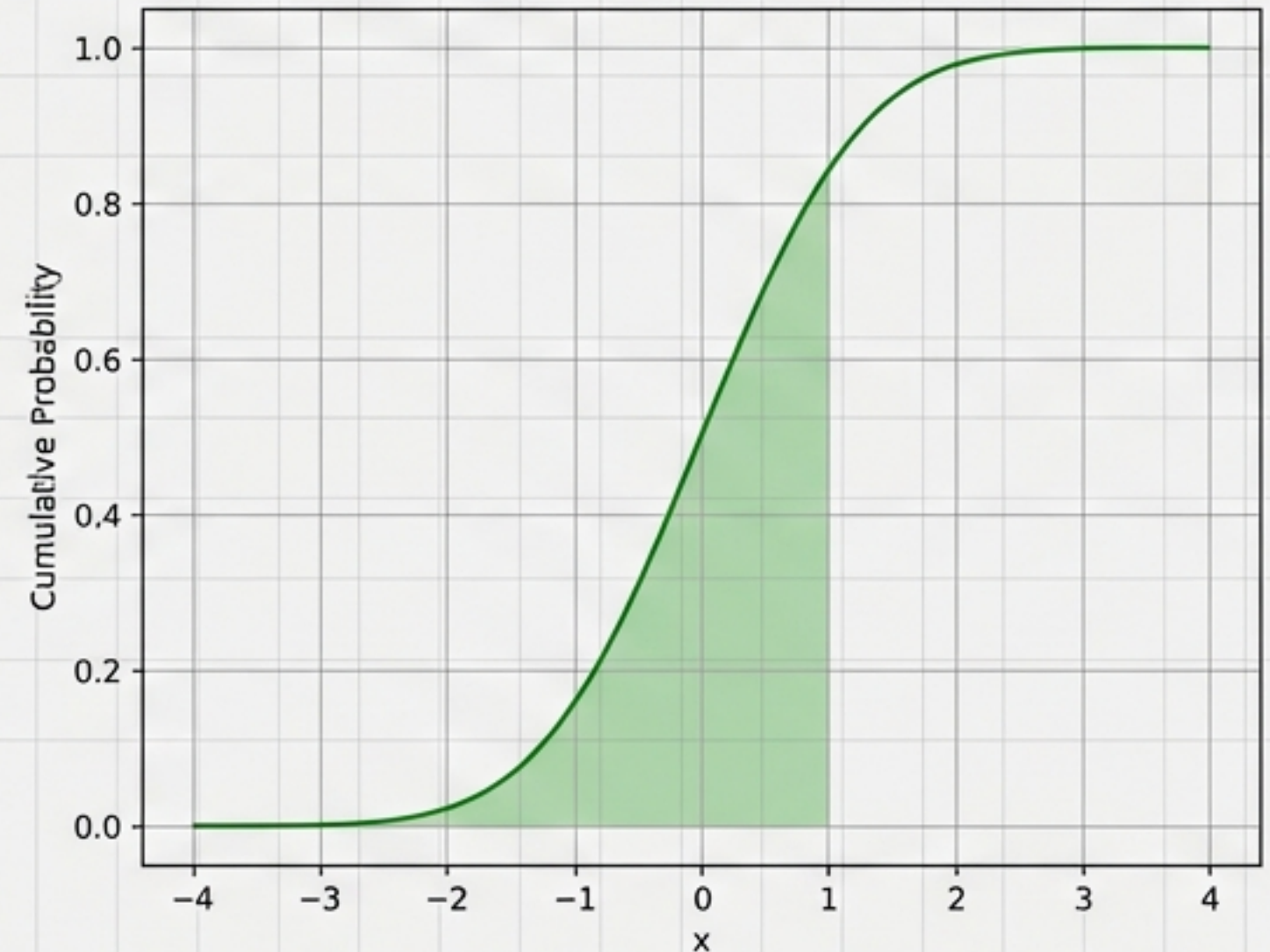
The CDF, denoted as $F(x)$, gives the probability that a random variable X is less than or equal to a value x .

$$F(x) = P(X \leq x)$$

It is obtained by integrating the PDF from negative infinity to x .

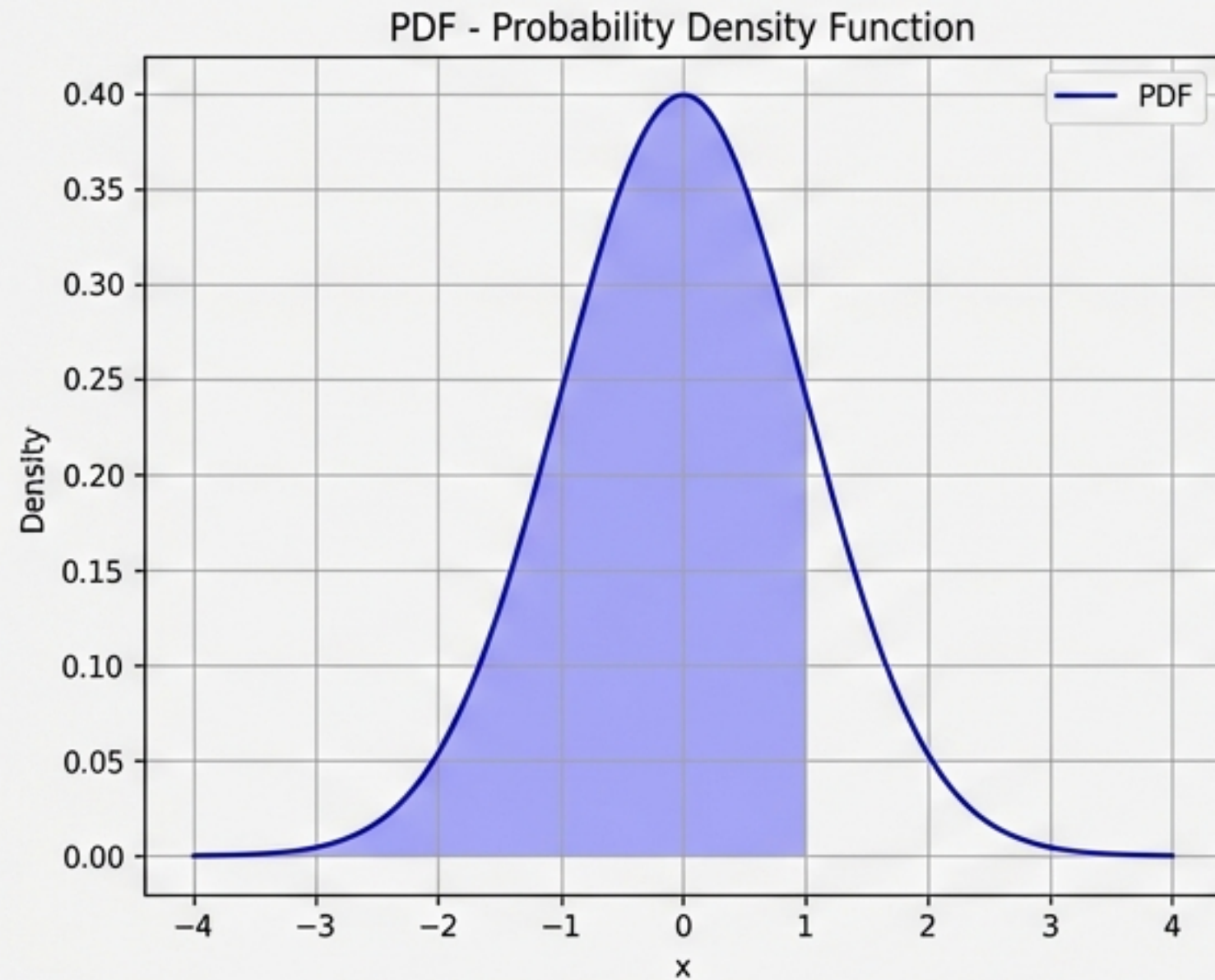
Visual Traits:

- An S-shaped curve (sigmoid-like).
- Non-decreasing function.
- Increases from $0 \rightarrow 1$.



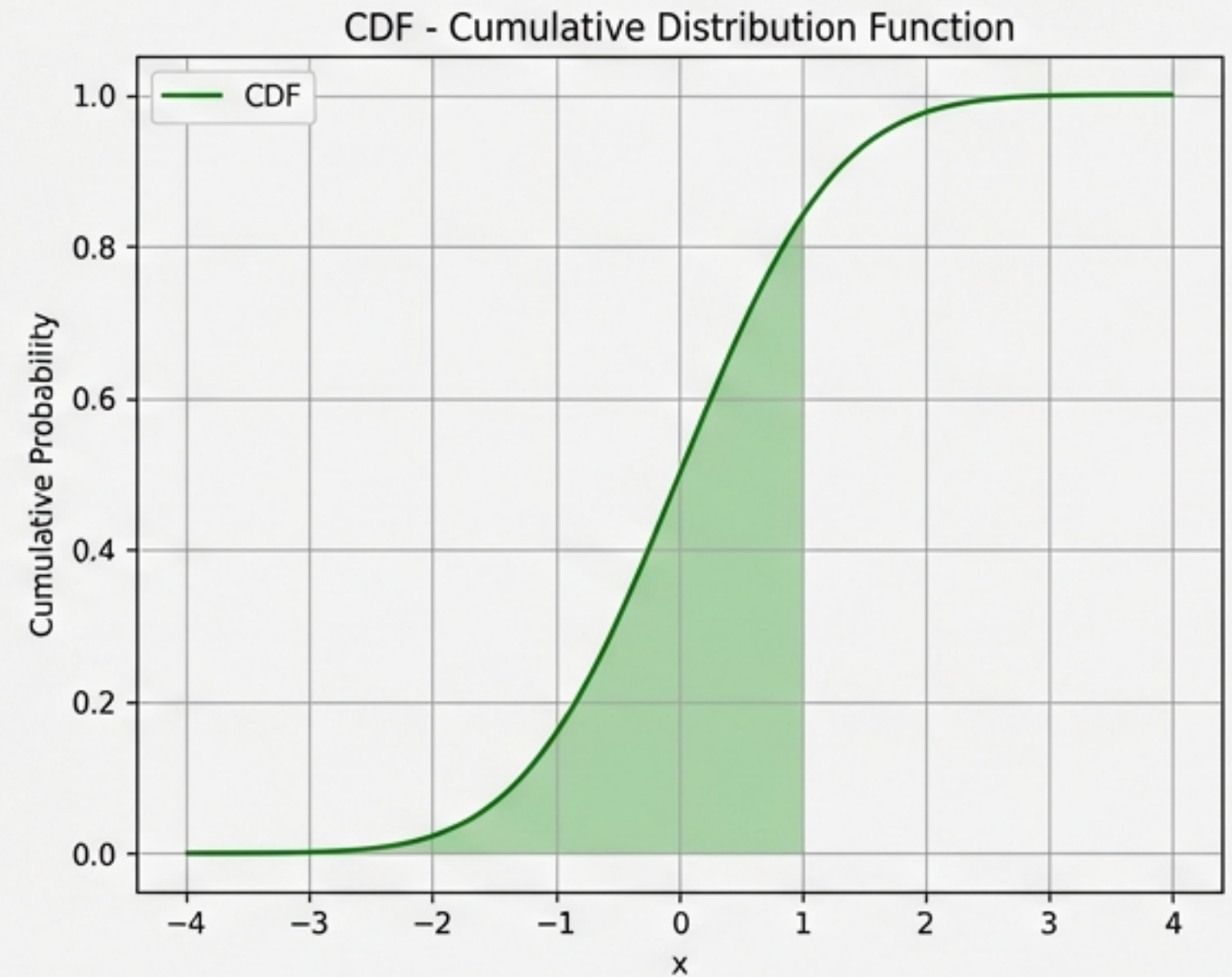
$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

The Visual Relationship: Height vs. Area



PDF describes HEIGHT.

Role: Defines the shape of the distribution and where data clusters.



CDF describes ACCUMULATED AREA.

Role: Calculates the total probability accumulated up to a specific point.

Key Takeaway: The PDF shows you the likelihood shape; the CDF tells you the total probability up to that point.

The Universal Ruler: The Standard Normal Distribution

To compare data points from different distributions, we standardize them into a common format. The Standard Normal Distribution (Z) is a special case where the Mean (μ) is 0 and the Standard Deviation (σ) is 1.

$$X \sim N(0, 1)$$

$$f(1) \approx 0.242$$

$$F(1) = P(Z \leq 1) \approx 0.8413$$

← Probability of being left of 1 standard deviation.

Measuring Distance with the Z-Score

A Z-score tells us exactly how many standard deviations a data point is away from the mean.

The diagram shows the Z-score formula with labels for each variable. The formula is $Z = \frac{X - \mu}{\sigma}$. The variable X is labeled "The data point you are analyzing". The variable μ is labeled "The mean of the population". The variable σ is labeled "The standard deviation". The variable Z is labeled "Standard Score".

$$Z = \frac{X - \mu}{\sigma}$$

Interpretation:

- Positive Z: The data point is above the mean.
- Negative Z: The data point is below the mean.

Applied Probability: The Single Threshold

Scenario:

Exam scores are normally distributed.

- Mean (μ) = 70
- Standard Deviation (σ) = 10
- Standard Deviation (σ) = 10

Question: What is the probability of a student scoring less than 85?

Solution Flow:

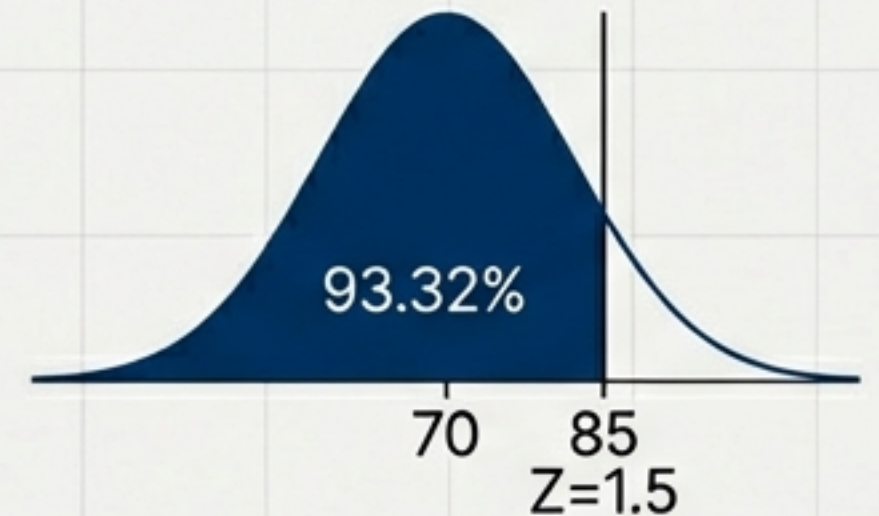
Step 1. Calculate Z

$$Z = \frac{85 - 70}{10} = 1.5$$

Step 2. Consult the Standard
Look up $P(Z \leq 1.5)$

Step 3. Result

0.9332



About 93.32% of students score below 85.

Applied Probability: The Interval

Same exam (Mean=70, SD=10).

Question: Probability of scoring between 60 and 80?

Calculate Lower Z (X=60):

$$Z = \frac{60 - 70}{10} = -1$$

Table value $\rightarrow$ 0.1587

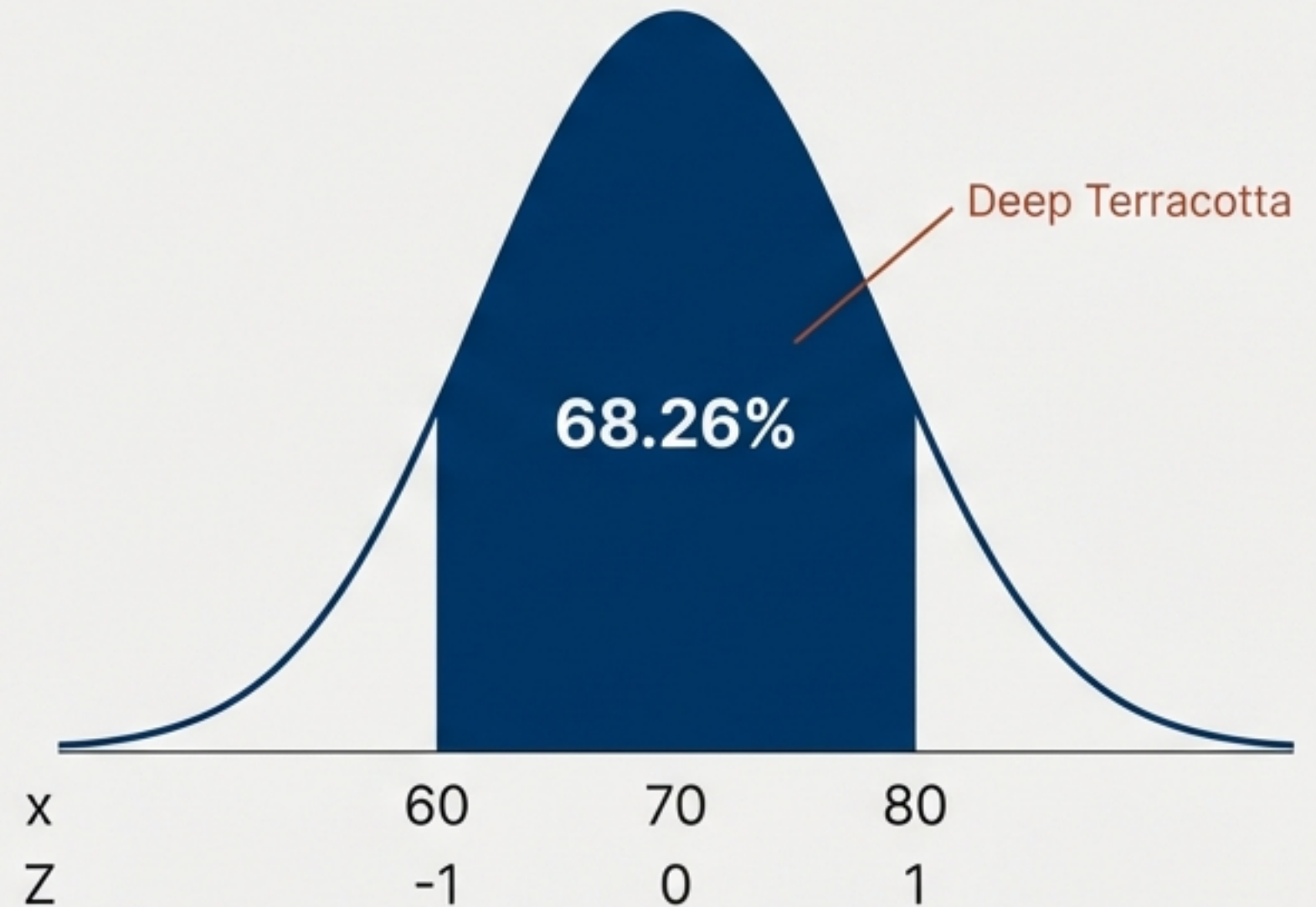
Calculate Upper Z (X=80):

$$Z = \frac{80 - 70}{10} = 1$$

Table value $\rightarrow$ 0.8413

Subtract:

$$P(60 \leq X \leq 80) = 0.8413 - 0.1587 = 0.6826$$



About 68.26% of students score between 60 and 80.

The Event Counter: The Poisson Distribution

When we stop measuring continuous variables and start counting events in a fixed interval, we need the Poisson Distribution.

Key Assumptions:

- Mean (Expected Value): $E[X] = \lambda$
 - Variance: $\text{Var}(X) = \lambda$
- 👉 The mean and variance are equal.

Real-World Examples:



Number of emails received in an hour.



Number of cars passing a toll booth per minute.



Number of printing errors per 100 pages.

The Poisson Formula

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

k = number of occurrences (0, 1, 2...)	λ = average rate of occurrence (mean)	$e \approx 2.718$ (Euler's constant)
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Key Statistic: In Poisson, the Mean equals the Variance.

$$E[X] = \text{Var}(X) = \lambda$$

Poisson in Practice: The Call Centre

Scenario:

A call centre receives an average of 5 calls per hour ($\lambda = 5$).

Question:

What is the probability of receiving exactly 3 calls in the next hour?

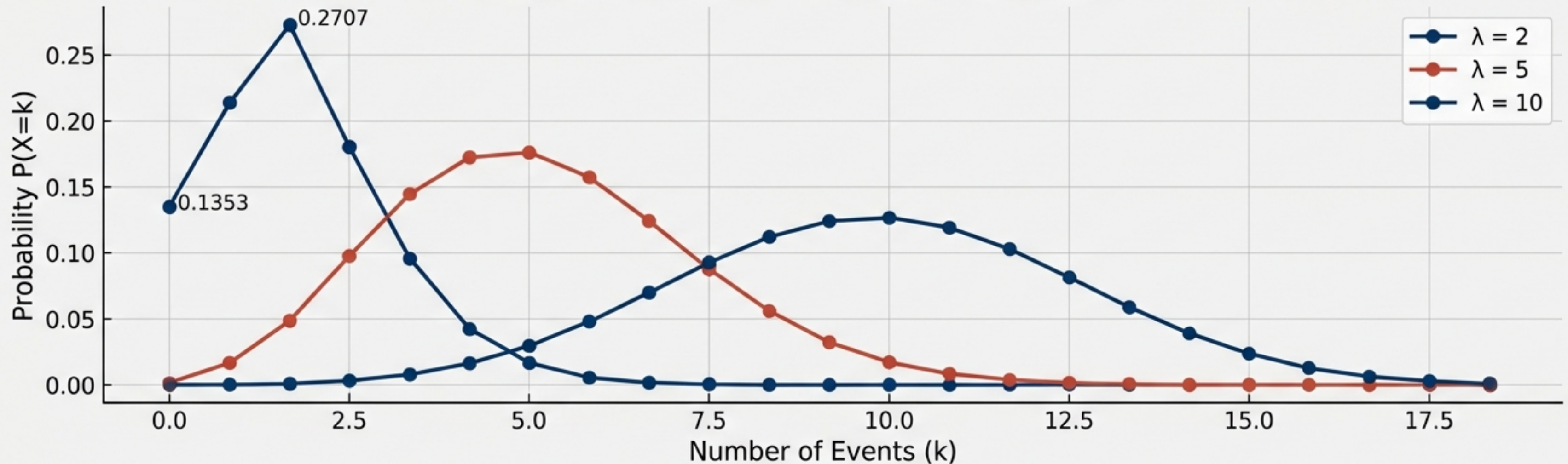
$$\begin{aligned} P(X = 3) &= \frac{e^{-5} \cdot 5^3}{3!} \\ &= \frac{0.0067 \cdot 125}{6} \\ &= 0.1404 \end{aligned}$$

There is a 14% chance of receiving exactly 3 calls, despite the average being 5.

The Power of Lambda (λ)

How the average rate changes the shape of probability.

Poisson Distribution for Different λ Values



Low λ (2): Probability mass is concentrated near zero; strongly skewed.

High λ (10): The distribution shifts right and becomes flatter (more spread out).

Takeaway: As the average rate (λ) increases, the distribution becomes more symmetric.

The Data Scientist's Lab: Python Implementation

Continuous - Normal Distribution

```
import scipy.stats as stats

# CDF: Prob of scoring < 85 (Mean 70, SD 10)
prob = stats.norm.cdf(85, loc=70, scale=10)
print(f'P(X < 85): {prob:.4f}')
```

Discrete - Poisson Distribution

```
from scipy.stats import poisson

# PMF: Prob of exactly 3 calls (Lambda 5)
prob = poisson.pmf(3, mu=5)
print(f'P(X=3): {prob:.4f}')
```


Choosing the Right Distribution

Distribution	Parameters	Primary Use Case
Normal (Gaussian)	μ (Mean), σ (Standard Deviation)	Determining probabilities over an interval (heights, scores, errors).
Poisson	λ (Rate)	Modeling rare events or arrival rates in a fixed interval (accidents, queues, word counts).

Correctly identifying your data type is the first step in modelling uncertainty.